Q14: Concepts, Features, and Memory

Directions, superposition, and where knowledge lives

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The question

Student question (#27)
How are “concepts” maintained and generated? Do they live in features? How do models reason and keep memory?

Concepts = directions, not neurons

- A concept $\approx$ a **direction** in activation space.
- One neuron participates in *many* concepts.
- Linear representation hypothesis: features are directions; operations approx. linear.

Superposition

- Networks pack **more features than dimensions**.
- Works when features are *sparse* (few active at once).
- Explains **polysemantic neurons**.
- Sparse autoencoders disentangle them back into meaningful features.

Where knowledge lives: MLP layers

- Feed-forward sublayers act as **key–value memories**.
- Causal tracing can localize — and *edit* — a specific fact.
- Attention *moves* info; MLPs *store* it.

Two kinds of memory

- **Weights**: long-term, learned in training, fixed at inference.
- **Context**: short-term working memory, gone when the chat ends.

A model doesn't "remember" past chats unless re-supplied / retrieved (Q15).